# PING needs gradient dampening, COBA does not

nb064 · 2 July 2026 · final · Gamma-gated sparsity

## Introduction

Surrogate-gradient training of the spiking network applies a voltage-gradient dampening factor (the *–v-grad-dampen* knob; theory in ar006) that scales down the gradient flowing through the leaky-integrate-and-fire membrane update. It is easy to read this as a generic BPTT stabiliser. This entry tests a sharper claim: dampening is specifically what the recurrent E→I→E loop needs, and the feedforward control does not need it at all.

Concretely — with $d$ the dampening factor (larger = more dampening; $d = 1$ is none):

- **COBA** (loop off, $\alpha_{EI} = 0$) trains to the same accuracy at $d = 1$ as at the canonical $d = 1000$.
- **PING** (loop on, $\alpha_{EI} = 1$) fails to train at $d = 1$ — its backpropagated gradient explodes through the inhibitory feedback and accuracy sits at chance — and only trains once dampening is applied.

If both halves hold, dampening is not a global trick but a loop-specific requirement, and the manuscript's §3-para-1 mechanism claim can be stated as a controlled result rather than a caveat.

## Method

**Grid.** Two architectures (COBA, PING) × a dampening ladder $d \in \{1, 10, 100, 1000\}$ × one seed = 8 networks. Each cell is otherwise the canonical nb025 recipe for its architecture; only $\alpha_{EI}$ (loop on/off, plus its matched $w_{\text{in}}$ / readout scale) and the swept $d$ differ. One seed suffices — the effect is a qualitative dissociation (chance vs converged), not an effect size. These cells are bespoke to this validation — the low-$d$ PING cells are deliberately unstable — so they are trained in this runner rather than in the nb022 shared hub.

**Run parameters.** Epochs are fixed (a convergence control, not a budget dial); the data budget is the runner's own declared scale. The scale reported is the one that actually produced the figures. Architectures: COBA (loop off) · PING (loop on); dampening ladder $d$: 1, 10, 100, 1000; seed 42; optimiser Adam, lr $4 \times 10^{-4}$; mem-mean readout.

**Readouts.** For each cell the runner reads best test accuracy (did it learn) and the per-epoch gradient norm (did BPTT stay bounded) from *metrics.json*. The predicted signature is a double dissociation: COBA's accuracy flat across the ladder while PING's rises from chance, and COBA's gradient norm bounded while PING's diverges at low $d$.

All network construction, training and metric logging happen in *cli.py*; the notebook only shells out and reads artifacts.

## Results

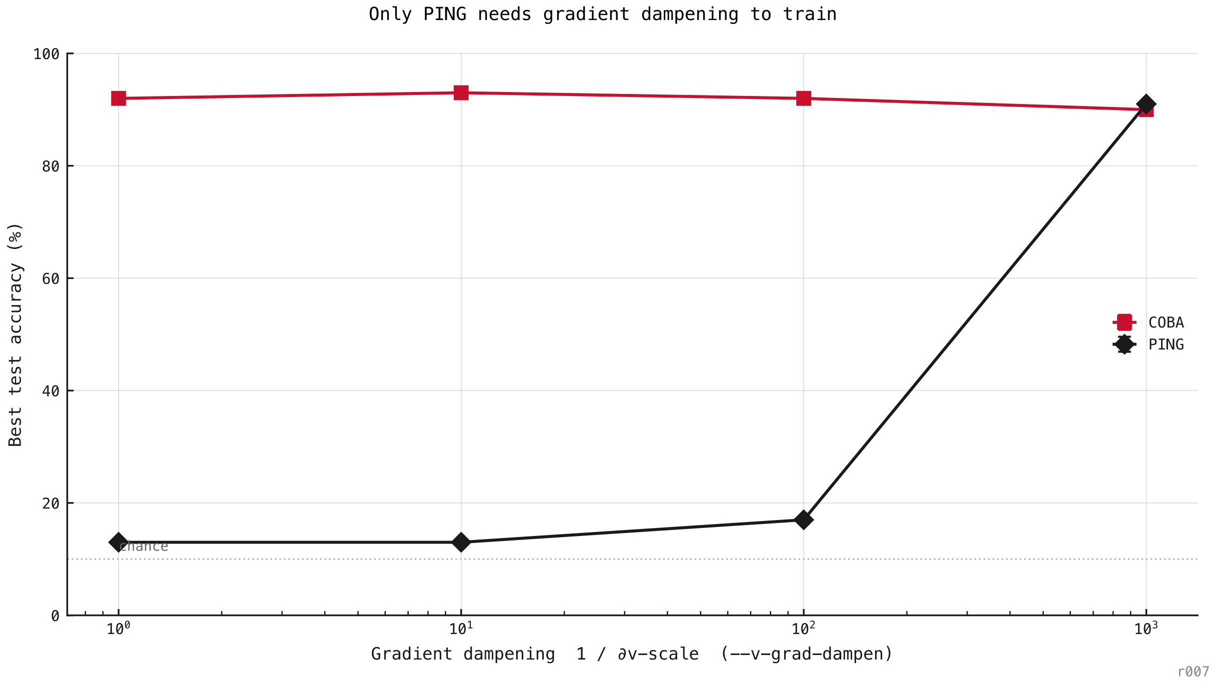


Figure 181: Best accuracy across the dampening ladder. COBA is flat — it trains at $d = 1$ as well as at $d = 1000$. PING climbs from chance, needing dampening to train.

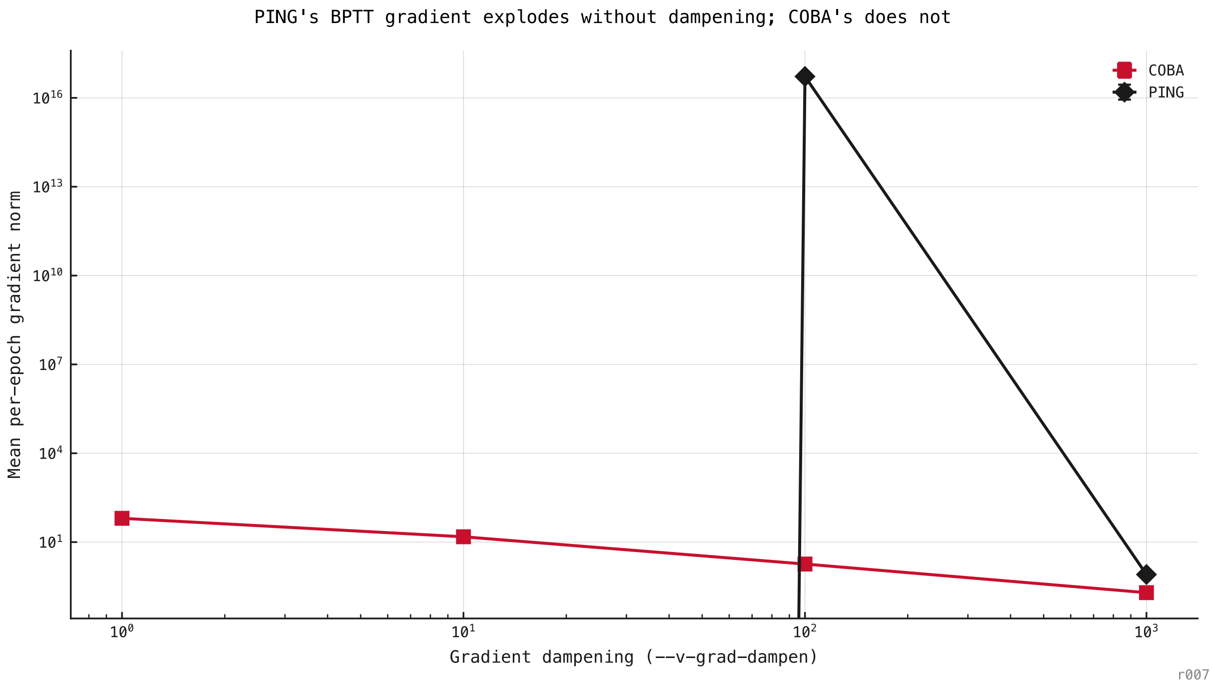


Figure 182: The mechanism: at low dampening PING's gradient norm blows up through the E→I→E loop, while COBA's stays bounded — which is why only PING's accuracy collapses.

## Discussion

If the dissociation holds, the E→I→E loop is the source of the training instability, not the spiking nonlinearity or depth per se — COBA shares both yet trains at $d = 1$. Dampening is therefore the price of the recurrent inhibitory feedback, and the manuscript can state the loop, not surrogate gradients in general, as what requires it.

## Success criteria

- **COBA does not need dampening**: COBA best accuracy at $d = 1$ is within 5 pp of its best across the ladder.
- **PING needs dampening**: PING best accuracy at $d = 1$ is within 15 pp of chance and more than 5 pp below PING at $d = 1000$.

Both are computed in the runner (*success* block of numbers.json); the claim is validated only if both hold.

## Next steps

If validated, fold the result into the ar010 item-1 write-up and upgrade the §3-para-1 manuscript claim from "is consistent with" to a controlled statement. If PING trains at $d = 1$ after all (claim fails), the dampening requirement is not loop-specific and the manuscript framing stays as-is.